Comparison with Published Studies

A study of Heckeberg, N. S. (2020) on the The systematics of the *Cervidae*: a total evidence approach, key phylogenetic findings include a confirmation of the two-subfamily classification: Cervinae (Muntiacini + Cervini) and Capreolinae (Alceini + Capreolini + Odocoileini + Rangiferini). All major clades (Cervinae, Cervini, Muntiacini, Capreolini, Odocoileini) recovered as monophyletic meaning that this group of organisms includes a common ancestor and all of its descendants. Heckeberg's results directly support the topology in the tree (revealed using FASTTREE). The two main clades (Cervinae and Capreolinae) also recovered with strong support. Within Cervinae, Muntiacus is sister to the Cervus clade, exactly as the tree shows. Within Capreolinae, *Capreolus* and *Odocoileus* are in separate tribes but grouped in the same subfamily. Heckeberg also specifically discusses Giraffa and Bovidae as outgroups within Ruminantia, which validate the choice of *Giraffa camelopardalis*.

On another study about Comparative genomics reveal phylogenetic relationship and chromosomal evolutionary events of eight Cervidae species, published by Tang, L., Dong, S., & Xing, X. in 2024, it uses chromosome-level whole-genome assemblies from eight Cervidae species (plus cattle as an outgroup) to resolve phylogeny and trace chromosomal evolution. The study finds phylogenetic tree built from 2,480 single-copy orthologous genes confirms two main branches: Capreolinae (reindeer) and Cervinae (Muntiacini + Cervini). And within Cervini, *Cervus canadensis* (wapiti) and *Cervus elaphus* (red deer) are sister taxa. Gene family evolution finds significant contraction of the olfactory receptor gene family across all eight Cervidae species, possibly related to habitat fragmentation, and significant expansion of the histone gene family which may have been involved in chromatin stability and gene regulation. Tang et al. (2024) provides whole-genome-level confirmation of the same two-subfamily topology the tree shows (recovered from FASTTREE). Their placement of *Cervus canadensis* and *Cervus elaphus* as sister species within Cervini matches the tree exactly. It also confirms *Muntiacus* as sister to Cervini within Cervinae.

Heckeberg included more deer species and fossil data, while Tang et al. used genome-wide data to better understand deer evolution. Both studies support the relationships shown in the phylogenetic tree.